

HDOC - Day 11 Activity

1. Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum of squares of its digits using a while loop. (E.g., Number: 123, Sum = 14)
2. Find the sum of cubes of its digits using a for loop. (E.g., Number: 123, Sum = 36)
3. Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop. (E.g., Number: 123, Difference = 22)

2. Algorithm

- **Step 1:** Start.
- **Step 2:** Display the menu choices to the user:
 - 1: Sum of squares of digits (using while loop)
 - 2: Sum of cubes of digits (using for loop)
 - 3: Difference between sum of cubes and sum of squares (using do-while loop)
- **Step 3:** Read the user's choice (choice).
- **Step 4:** Read a positive integer (num).
- **Step 5:** Use switch(choice):
 - **If Choice == 1:**
 - Initialize sum = 0, temp = num.
 - While temp > 0:
 - Extract last digit: $\text{rem} = \text{temp} \% 10$
 - Add square to sum: $\text{sum} = \text{sum} + (\text{rem} * \text{rem})$
 - Remove last digit: $\text{temp} = \text{temp} / 10$
 - Print sum.

- **If Choice == 2:**
 - Initialize sum = 0.
 - For temp = num; temp > 0; temp = temp / 10:
 - Extract last digit: rem = temp % 10
 - Add cube to sum: sum = sum + (rem * rem * rem)
 - Print sum.
- **If Choice == 3:**
 - Initialize sumSq = 0, sumCu = 0, temp = num.
 - Do:
 - Extract last digit: rem = temp % 10
 - sumSq = sumSq + (rem * rem)
 - sumCu = sumCu + (rem * rem * rem)
 - temp = temp / 10
 - While temp > 0.
 - Calculate diff = sumCu - sumSq.
 - Print diff.
- **Default:**
 - Print "Invalid Choice".
- **Step 6:** Stop.

3. Test Case Table

Test case	choice	Input num	Expected output	Actual op	status
1.	1	456	Sum=77	Sum=77	pass
2.	2	456	Sum=405	Sum = 405	pass
3.	3	456	Difference=328	Difference=328	pass

4. C Program Source Code

```
UW PICO 5.09 File: 11q1.c

#include <stdio.h>

int main() {
    int choice, num, temp, rem;
    int sum = 0;
    int sumSq = 0, sumCu = 0, diff = 0;

    printf("---- MENU ----\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference between sum of cubes and sum of squares (do-while loop)\n");
    printf("Enter your choice (1-3): ");
    scanf("%d", &choice);

    if (choice >= 1 && choice <= 3) {
        printf("Enter a positive integer: ");
        scanf("%d", &num);
    }

    switch (choice) {
        case 1:
            // Case 1: Sum of squares using while loop
            temp = num;
            sum = 0;
            while (temp > 0) {
                rem = temp % 10;
                sum = sum + (rem * rem);
                temp = temp / 10;
            }
            printf("Sum of squares of digits = %d\n", sum);
            break;

        case 2:
            // Case 2: Sum of cubes using for loop
            sum = 0;
            for (temp = num; temp > 0; temp = temp / 10) {
                rem = temp % 10;
                sum = sum + (rem * rem * rem);
            }
            printf("Sum of cubes of digits = %d\n", sum);
            break;

        case 3:
            // Case 3: Difference using do-while loop
            temp = num;
            sumSq = 0;
            sumCu = 0;
            do {
                rem = temp % 10;
                sumSq = sumSq + (rem * rem);
                sumCu = sumCu + (rem * rem * rem);
                temp = temp / 10;
            } while (temp > 0);
    }
}
```

```

        diff = sumCu - sumSq;
        printf("Difference = %d\n", diff);
        break;

    default:
        printf("Invalid choice! Please select 1, 2, or 3.\n");
    }

    return 0;
}

```

- EXECUTION AND OUTPUT :

```

[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./11Q1
--- MENU ---
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice (1-3): 1
Enter a positive integer: 456
Sum of squares of digits = 77
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./11Q1
--- MENU ---
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice (1-3): 2
Enter a positive integer: 456
Sum of cubes of digits = 405
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./11Q1
--- MENU ---
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice (1-3): 3
Enter a positive integer: 456
Difference = 328
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %

```